Real-Time Multimodal Social Media Harm Detection and Reporting System

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Introduction

Social media platforms generate massive amounts of user content every second, including text posts, comments, images, videos, and live-stream media. Along with the benefits of digital communication, these platforms also face serious challenges such as cyberbullying, hate speech, violent content, fake media, and abusive online behavior.

Traditional moderation systems are often unable to handle large-scale real-time content efficiently. Therefore, intelligent AI-based moderation systems are required to automatically detect and report harmful activities across multiple forms of media.

Problem Statement

The rapid growth of social media platforms has increased the spread of harmful and inappropriate content across text, images, videos, and live-stream data. Manual moderation approaches are slow, expensive, and unable to process large volumes of multimodal content in real time.

This project aims to develop a Real-Time Multimodal Social Media Harm Detection and Reporting System using Artificial Intelligence and MLOps practices. The system will automatically analyze social media content to detect harmful activities such as abusive language, violent media, suspicious content, and offensive behavior. It will generate automated alerts and reports while supporting scalable deployment and monitoring for real-time moderation.

Project Ideation

The proposed system is inspired by modern AI moderation systems used by large social media platforms. The idea is to create a smart moderation pipeline capable of processing and analyzing multiple media formats in real time.

The platform will use Natural Language Processing (NLP) to analyze text-based content and Computer Vision (CV) techniques to analyze images and video frames. For live-stream monitoring, the system will process frames periodically and identify harmful or suspicious activities instantly.

The project will also integrate MLOps concepts such as model deployment, experiment tracking, monitoring, and containerization to create a scalable and production-oriented AI system.